Paleomagnetism and Secular Variation of Easter Island Basalts

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Sixty-five Brunhes age lava flows from Easter Island are analyzed for secular variation statistics and mean geomagnetic field parameters. The present rotation axis is enclosed within the 95% confidence limits of the mean virtual geomagnetic pole (θ = 87.4°, ϕ = 204.2°; dm = 4.13°, dp = 2.57°). The angular standard deviation of a subset of 51 poles is 12.8° with upper and lower confidence limits of 14.8° and 11.3°. The secular variation of the young Easter Island basalts lies above a pure dipole wobble component of secular variation but below the lower limit for all proposed models. Thus it appears that the Pacific secular variation low extends into the southeast Pacific but is not as pronounced as it is around Hawaii.

INTRODUCTION

Both geomagnetic observatory data and paleomagnetic data provide strong evidence for a region of anomalously low secular variation due to a subdued nondipole geomagnetic field in the central Pacific [*Cox and Doell*, 1964; *Doell and Cox*, 1971, 1972]. In particular, Brunhes epoch basalts from the Hawaiian Islands give anomalously low values of angular dispersion when they are compared to similar Brunhes age paleomagnetic data from other areas [e.g., *Doell*, 1972*a*; *Doell and Cox*, 1972]. Although the existence of the Pacific secular variation low is well documented for the last few million years in Hawaii [*Doell*, 1972*b*; *Doell and Dalrymple*, 1973], the lateral extent and persistence of the low throughout the Pacific region are not well known. This study of basalts from Easter Island was undertaken in part to investigate the secular variation values for the southeastern Pacific region.

Geologic setting. Easter Island is a small isolated outcrop of volcanic rock in the southeast Pacific (27.1°S, 109.2°W) located 1380 km west of Chile, the nearest continental land mass. The island rises from the Sala y Gomez ridge just east of the axis of the east Pacific rise. It is part of a chain of seamounts interpreted to be related to a 'hot spot' or 'mantle plume' generated during the last 10 m.y. within the Nazca lithospheric plate [*Herron*, 1972*a*]. Although well known as an archeological site, only cursory geological investigations have been made of the island [*Chubb*, 1933; *Bandy*, 1937; *Baker*, 1966]. All the studies recognized the existence of three distinct episodes of volcanism and named each for its major volcano: Poike, Rano Kau, and Terevaka. *Baker* [1966] determined the relative ages of events and classified Poike and Rano Kau as the older episodes with Terevaka, covering the major part of the island, as the youngest event (Figure 1). An extensive number of once-active volcanic vents and pyroclastic centers, especially during the Terevaka episode, make it difficult to correlate individual flows across the island or to further delineate the volcanic stratigraphy. The lava flows are transitional between island tholeiites and alkali basalts with even the more basic rocks saturated in silica (J. Clark, personal communication, 1975). Easter Island is situated between marine magnetic anomalies 2 and 3 [*Herron*, 1972*b*], and thus the sea floor in the vicinity of the island has a minimum age of 2 m.y. and a maximum age of 5 m.y. As part of this study, potassium-argon age dates have been obtained for 30 lava flows. Preliminary results indicate an age of 2.5 m.y. for Poike lavas, 1.0 m.y. for Rano Kau lavas, and an age less than 0.24 m.y. for the Terevaka flows [*Clark and Dymond*, 1974]. The 65 flows analyzed for secular variation values are all from the Terevaka episode and are definitely Brunhes in age.

Field and laboratory methods. Owing to the failure of the portable drill to arrive for the initial phase of the field work, only hand samples (1–3) were collected from 25 flows in reconnaissance sampling for geochemical and future paleomagnetic studies. With limited time and field logistics support for detailed sampling, the sampling with the portable drill [*Doell and Cox*, 1965*a*] was concentrated along the accessible sea cliffs formed by the younger volcanic sequence (Figure 1). Sixty-five Terevaka episode lava flows were drilled along with four Poike episode flows. The usual sampling technique consisted of drilling four cores per flow and determining the geographic orientation prior to breaking the sample free from the host rock. In five cases, only three usable samples were obtained. Each core was cut into two or more specimens.

The natural remanent magnetization (NRM) of all the specimens was measured on a flux gate spinner magnetometer [*Isaacson*, 1974]. A four-axis tumbler with an alternating magnetic field was used for demagnetization experiments [*Doell and Cox*, 1965*a*]. Two or more specimens from each flow were subjected to progressive step demagnetization at field levels of 25, 50, 100, 200, 400, and 800 Oe. The optimum demagnetization level for each flow was that at which the test specimens were most closely clustered. Alpha 95 values of less than 10° were desired, and in a few instances entire flows were demagnetized at several levels to find the best cluster of directions. Approximately two thirds of the 65 Terevaka flows had well-clustered NRM directions which moved very little on demagnetization. The remaining flows had at least one specimen with a direction of magnetization removed from the main grouping and were demagnetized at levels up to 200 Oe to remove unstable secondary components of magnetization. A typical step demagnetization curve and direction plot for one such flow (flow 23) is shown in Figure 2.

MEASUREMENTS AND RESULTS

A detailed investigation of the geochemistry and geochronology of the island is in progress [*Clark and Dymond*, 1974]. The 30 lava flows presently dated by the potassium-argon

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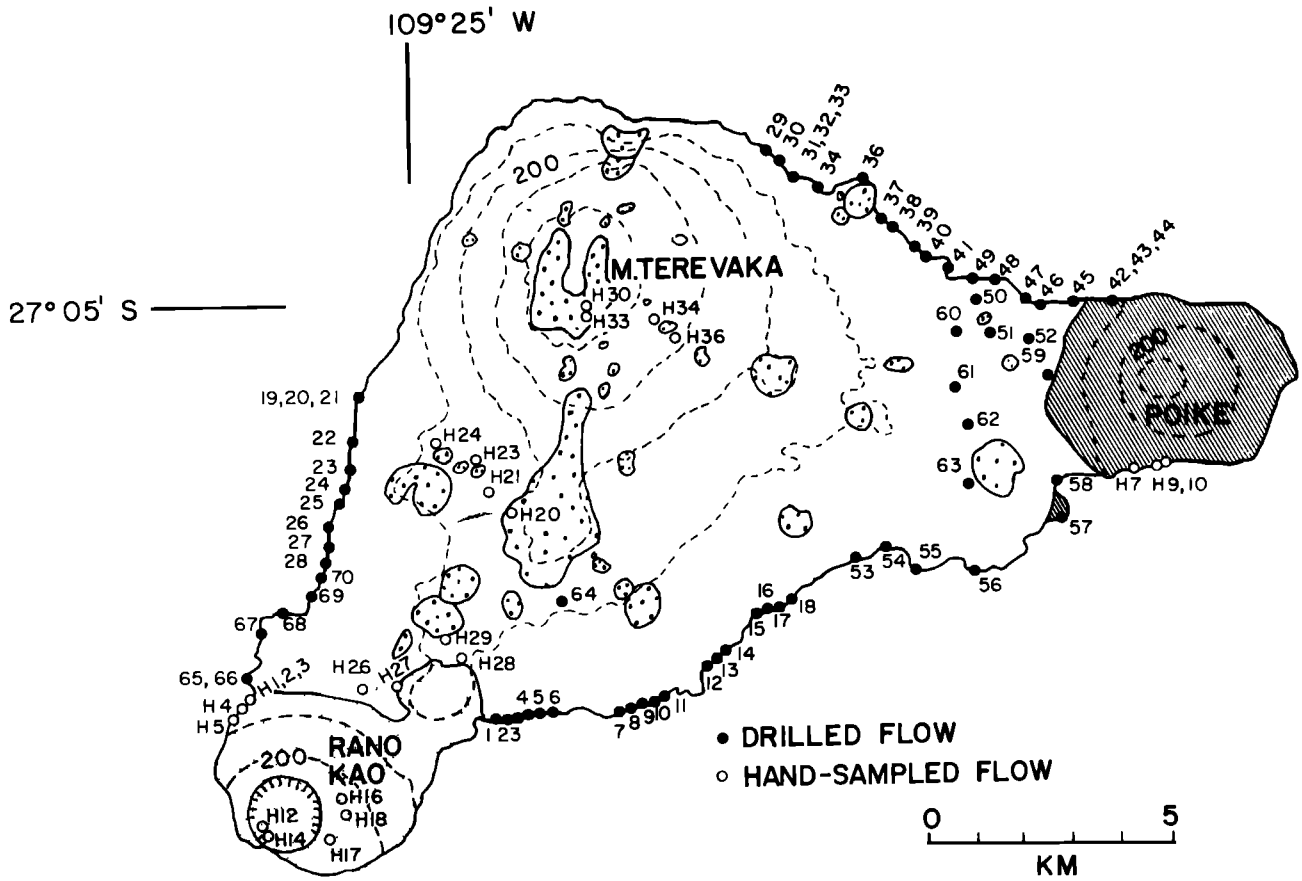


Fig. 1. Geology and sample locations, Easter Island. Geology is simplified from *Baker* [1966]. The white area is Terevaka episode volcanics; the areas with large dots are associated pyroclastic centers. Contour interval is 100 m.

method readily identify the three episodes of volcanism recognized in field studies. Poike, the easternmost volcano, is the oldest, with an age of 2.5 ± 0.2 m.y. based upon four determinations. Five of the flows from the southern volcano Rano Kau have been dated, all with ages of approximately 1.0 m.y. Terevaka is definitely the youngest episode on the island; the 15 dated flows all have ages of 240,000 yr or less. The three volcanic episodes on Easter Island fall into the three most recent polarity epochs. The Poike volcanics lie in the Gauss normal epoch, the Rano Kau lavas are in the Matuyama reversed epoch, and the Terevaka flows are in the youngest epoch, the Brunhes normal.

The mean direction of magnetization, virtual geomagnetic pole (VGP) position, and associated precision parameters for the drilled Terevaka flows are listed in Table 1. Two sites, flows 5 and 60, are not included, as random directions were

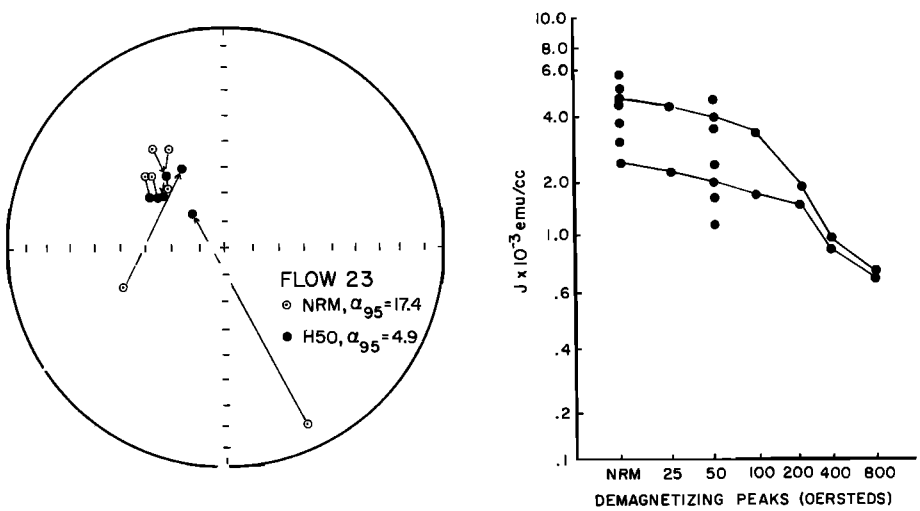


Fig. 2. Demagnetization experiments on flow 23. The left diagram is a stereoprojection of paleomagnetic directions for the seven specimens of flow 23. Open circles with dots are NRM; black circles are after demagnetization at 50 Oe. The right diagram is a plot of magnetic intensity versus demagnetizing level for the same seven specimens, where two specimens have been step-demagnetized to 800 Oe.

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TABLE 1. Paleomagnetic Data From Easter Island Terevaka Lava Flows

Unit	N	I, deg	D, deg	R	K	α_{95}	θ, deg	ϕ, deg	EXP
1	3	−34.2	32.4	2.9914	232	8.1	59.1	331.4	NRM
2	4	−38.6	26.8	3.9859	213	6.3	65.1	333.5	NRM
3	4	−43.0	356.3	3.8279	17	22.6	86.0	192.9	H200
4	4	−49.3	0.0	3.9799	149	7.6	87.0	70.8	NRM
6	4	−34.8	348.9	3.9819	165	7.2	77.0	195.9	NRM
7	4	−43.5	5.3	3.9839	186	6.8	84.9	320.5	H50
8	4	−42.3	10.5	3.9808	156	7.4	80.2	326.5	H50
9	4	−59.3	329.5	3.9881	48	13.3	61.7	125.3	H150
10	4	−54.5	339.1	3.9498	60	12.0	70.5	131.3	H100
11	4	−52.6	11.6	3.9790	143	7.7	78.3	13.9	NRM
12	3	−46.1	4.4	2.9388	33	21.9	86.1	345.5	H50
13	3	−42.3	0.2	2.9924	263	7.6	87.3	254.1	H50
14	4	−53.8	306.1	3.9253	40	14.7	43.7	137.4	H200
15	3	−36.6	12.8	2.9671	61	16.0	76.5	313.1	H200
16	4	−35.3	4.9	3.9913	344	5.0	81.1	281.9	NRM
17	4	−44.1	16.6	3.9494	59	12.0	75.1	339.0	NRM
18	3	−47.4	3.5	2.9994	3258	2.2	86.6	4.7	NRM
19	4	−46.6	3.8	3.9705	101	9.2	86.5	353.3	NRM
20	4	−44.0	5.6	3.9906	318	5.2	84.8	327.0	NRM
21	3	−53.1	354.8	2.9991	2130	2.7	82.1	103.6	NRM
22	4	−57.8	329.3	3.9930	426	4.5	61.9	128.5	NRM
23	4	−57.6	325.7	3.9915	352	4.9	59.3	130.2	H50
24	3	−46.6	5.7	2.9935	308	7.0	84.9	349.5	NRM
25	4	−40.7	14.9	3.9895	287	5.4	76.0	327.4	NRM
26	4	−56.3	324.3	3.9447	54	12.6	58.4	133.1	NRM
27	4	−44.7	340.6	3.9416	51	12.9	72.7	158.3	NRM
28	4	−49.5	338.2	3.9550	67	11.3	70.6	145.5	NRM
29	4	−41.2	2.6	3.9921	380	4.7	85.9	285.1	NRM
30	4	−48.9	357.9	3.9856	208	6.4	86.7	103.6	NRM
31	4	−43.1	2.6	3.9542	66	11.4	86.9	300.1	NRM
32	3	−62.1	312.6	2.9975	784	4.4	48.7	124.3	H200
33	4	−62.5	326.2	3.9865	222	6.2	58.1	119.7	NRM
34	4	−68.7	321.2	3.8440	19	21.5	51.8	108.8	H50
36	4	−37.4	10.0	3.9932	443	4.4	79.0	308.4	NRM
37	4	−45.8	14.1	3.9935	462	4.3	77.5	344.1	NRM
38	4	−48.1	352.2	3.9440	54	12.7	82.8	141.5	H50
39	4	−37.4	8.6	3.9898	293	5.4	80.1	304.0	NRM
40	4	−47.1	6.3	3.9831	177	6.9	84.3	354.2	NRM
41	4	−42.0	357.0	3.9878	246	5.9	86.1	206.5	NRM
45	4	−35.0	3.1	3.9759	125	8.3	81.7	271.2	H100
46	3	−35.6	358.3	2.9939	327	6.8	82.4	238.0	H100
47	4	−31.7	6.1	3.9753	122	8.4	78.6	281.4	NRM
48	4	−40.6	356.9	3.9812	159	7.3	85.2	214.1	H100
49	4	−30.4	27.0	3.9200	38	15.2	62.8	322.7	NRM
50	3	−30.8	345.2	2.8729	16	32.2	72.8	194.6	H400
51	4	−25.7	337.0	3.9050	32	16.6	64.6	188.1	H50
52	3	−42.8	333.7	2.9335	30	22.9	66.3	159.8	H50
53	3	−55.2	353.9	2.9824	114	11.6	80.0	100.0	H50
54	4	−16.8	344.4	3.3487	5	48.2	66.3	208.9	H100
55	4	−52.4	17.4	3.9496	60	12.0	73.9	5.6	NRM
56	4	−39.5	5.0	3.9309	43	14.1	83.5	294.6	NRM
58	4	−37.2	15.9	3.9314	44	14.1	74.2	320.1	NRM
59	4	−42.4	11.0	3.9408	51	13.0	79.7	328.1	NRM
60	4	−39.9	344.8	3.7861	14	25.4	75.6	174.8	H100
62	4	−42.6	359.9	3.9658	88	9.9	87.5	247.2	H50
63	4	−44.2	1.4	3.9884	257	5.7	88.3	295.5	NRM
64	4	−49.7	12.2	3.9936	467	4.3	78.3	0.8	NRM
65	4	−43.3	355.1	3.9806	155	7.4	86.2	182.7	NRM
66	4	−36.9	352.0	3.9658	88	9.9	80.0	200.4	NRM
67	4	−41.5	1.7	3.9884	258	5.7	86.4	274.8	NRM
68	4	−36.5	348.6	3.9952	627	3.7	77.5	191.1	NRM
69	4	−48.9	5.8	3.9716	106	9.0	84.2	8.9	NRM
70	4	−37.6	348.8	3.7966	15	24.8	78.1	188.7	H200
Mean	63	−43.4	357.9	61.8714	30	3.3	87.4	204.2	
Mean 2	54	−44.4	0.3	52.4046	33	3.4	88.9	264.4	
Set A	51	−43.2	358.6	49.4819	33	3.5	87.7	218.3	
Set B	31	−43.3	1.4	30.4007	50	3.7	87.7	284.5	

N represents number of separate cores per flow; I, inclination below the horizontal; D, declination; R, length of the resultant vector; K, precision parameter; α_{95}, radius of the cone of 95% confidence about the resultant vector; θ, latitude of the VGP; ϕ, longitude of the VGP; and EXP, level at which remanent magnetization was measured.

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obtained for these flows. The standard methods described by *Fisher* [1953] were used for the computations.

The directions of magnetization for the hand samples which were cored later in the laboratory are not included. A jury-rigged field-orienting device, limited sampling, and additional errors from reorienting the samples in the laboratory contribute to increased scatter for these results. It should be noted that all the flows have a normal polarity except for flow H12, which has a reversed polarity based upon one sample.

Based upon the age control, normal polarities are expected for the Terevaka and Poike flows; reversed polarities are expected for the Rano Kau lavas, yet 9 of the 10 flows sampled have a normal polarity. The precision of the age determinations is such that it is possible that the Jaramillo normal event has been sampled (J. Clark, personal communication, 1975). However, *Watkins* [1968] and *Watkins and Abdel-Monem* [1971] suggest that an additional event may exist at 1.07 m.y.; this event, if real, may have been sampled.

The VGP positions for 63 Brunhes age flows are plotted in Figure 3. Included is the mean pole position of 67.5°N and 55.9°W surrounded by an oval of confidence at the 95% level with major axis of 4.13° and minor axis of 2.57°. The statistical oval just includes the present rotational pole, as is expected for rocks this young. *Cox and Doell* [1960], *Irving* [1964], and more recently *Opdyke and Henry* [1969] have noted that the mean poles for upper Tertiary to Recent rocks are very close to the geographic pole. It appears that for the time spanned by these lava flows, the youngest third of the Brunhes epoch, the earth's field was that of a geocentric axial dipole.

Although the VGP's cluster around the present rotational axis, there is a noticeable scatter of eight poles between 90°E and 150°E longitude. These poles have a minimum polar latitude of 43°N. Three of the flows (flows 32, 33, and 34) are found in sequence in a sea cliff on the north side of the island. Exact stratigraphic relationships between the other flows with displaced VGP's are not known, but both the south side (flows 9 and 14) and west side (flows 22, 23, and 26) of the island are represented.

Departures of the geomagnetic field from its predominant polar position, in the context of complete reversals or events or in aborted reversals or excursions, have been discussed by many [e.g., *Cox et al.*, 1964; *McElhinny*, 1973]. Several possible events have been reported for Brunhes age rocks [*Bonhommet and Zahringer*, 1969; *Smith and Foster*, 1969; *Morner et al.*, 1971], together with various excursions [*Watkins and Nougier*, 1973; *Barbetti and McElhinny*, 1972].

The data from Easter Island add to this list of possible field excursions in the Brunhes, although the excursion cannot be followed entirely, due to the undetermined relationships between the lava flows. One of the flows with a low VGP latitude was dated by potassium-argon at 127,000 ± 58,460 yr (J. Clark, personal communication, 1975) and may be related to the Blake event [*Smith and Foster*, 1969; *Wollin et al.*, 1971] dated between 108,000 and 114,000 yr.

Figure 4 is a plot of field directions for 54 of the measured

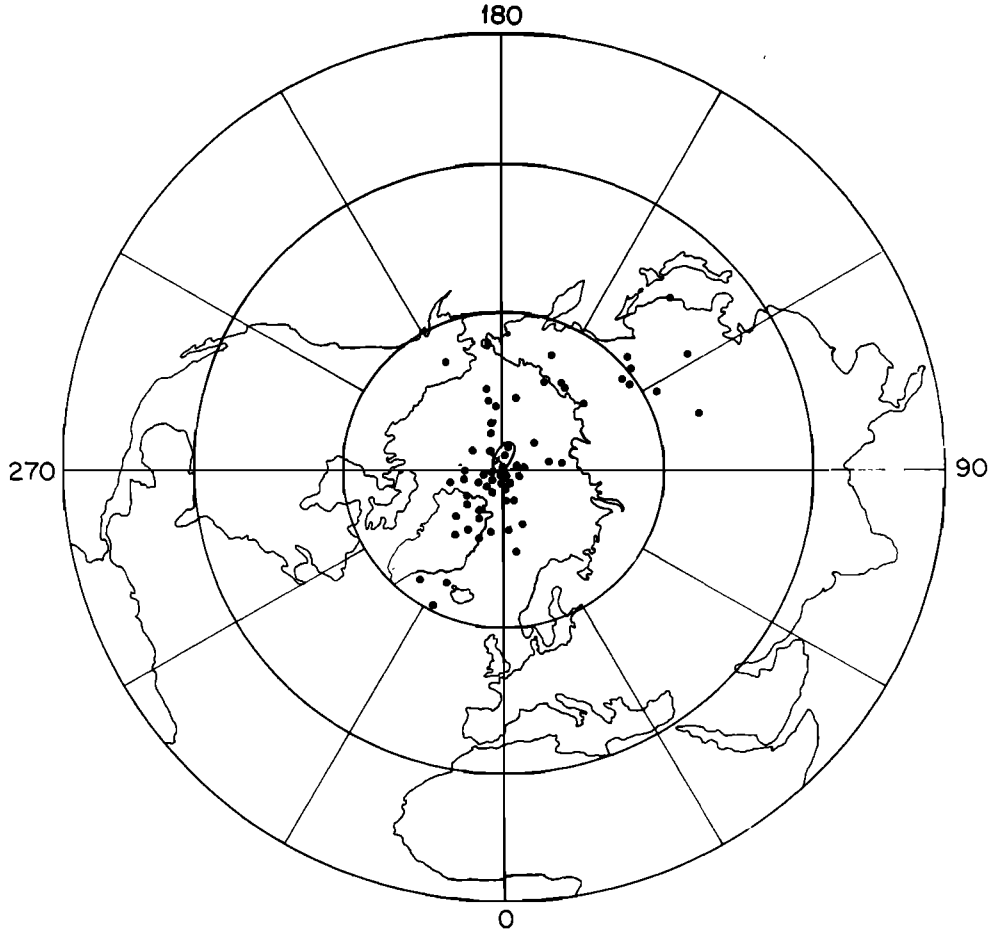


Fig. 3. VGP positions for all the Terevaka lava flows. The small cross in the center is the mean VGP with its oval of confidence at the 95% level. Exact data are given in Table 1.

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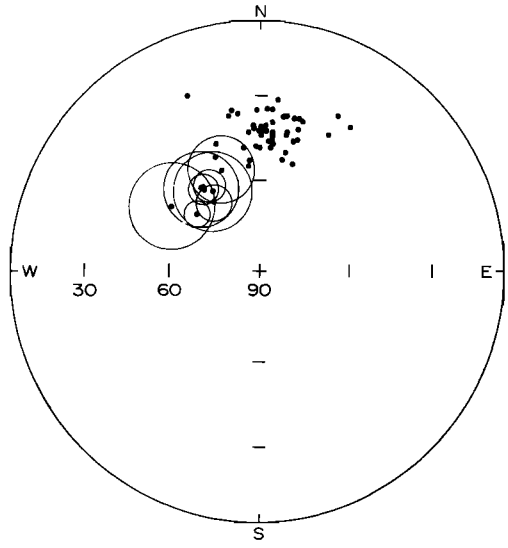


Fig. 4. Field directions for 54 Terevaka flows with α_{95} less than 16°. Circles are 95% confidence circles about the 10 flows with displaced VGP's. Black dots represent normal directions with negative inclinations.

flows. This subset (mean 2) includes all flows with α_{95} values less than 16°, based on the histogram of all α_{95} values (Figure 5). The flows with scattered VGP's are shown with their respective circles of 95% confidence. It can be seen that the flows in question are displaced from the main body of data and represent a steepening of inclination and a more westerly declination. If related to the Blake event, only a small part of the event has been recorded here. This would not be surprising, as the episodic nature of volcanism often leaves large gaps in the record. On the other hand, these displaced directions may be related to swings of the nondipole field, as observed on Hawaii [*Doell and Cox*, 1965*b*], although these swings are usually toward shallower rather than steeper inclinations.

Secular variation. The secular variation of the magnetic field for one location is best expressed as an angular standard deviation of paleomagnetic directions or VGP's. This deviation must be corrected for errors due to within-site dispersion and the possible presence of local anomalies at the location. A method for calculating the total angular standard deviation (S_T) has been introduced by *Cox and Doell* [1964]. Subsequent papers by these authors [*Cox*, 1969*a*, 1971; *Doell*, 1970] have developed techniques to handle the possible errors in determining the deviation. A synthesis of these statistical methods has been presented by *Ellwood et al.* [1973], and details and equations will not be repeated here.

It is important to note that the angular standard deviation can be calculated either for directions or poles. *Doell* [1970] and *Doell and Cox* [1972] give convincing arguments for the latter together with computation with respect to the present geographic pole. Since most studies of Brunhes age rocks yield mean poles not statistically different from the rotation axis, this has become the preferred procedure.

Easter Island Secular Variation

An important consideration in determining secular variation values is the selection of data to be used. Criteria established by Cox and Doell [*Cox*, 1969*a*; *Doell*, 1970; *Doell and Cox*, 1972] reject any data with α_{95} values greater than 9° and/or a VGP latitude of less than 46°. These investigators also require six to eight separate samples per flow. Other authors [*Ellwood et al.*, 1973; *Watkins and Nougier*, 1973] discard flows with low VGP latitudes but make no rejections on α_{95} values. A case has been made for as few as four sites per flow providing adequate results [*Ellwood et al.*, 1973].

Figure 5 is a histogram of the α_{95} values for the 63 Terevaka flows. The great majority of the flows fall between 2° and 16°, only 9 flows having α_{95} values greater than 16°. Two sets of statistics were calculated for the Easter Island data. One set (set A) contained only flows with α_{95} less than 10°. The other set (set B) included all the flows regardless of α_{95} values. For both sets the flows with VGP latitudes less than 46° were removed, as were any flows with less than four separate sample sites. This left 31 flows for set A and 51 flows for set B.

With the use of the Cox and Doell statistical techniques the angular standard deviations of the VGP's with respect to both the geographic pole and the mean VGP were calculated for each set and are presented in Table 2. The final measure of deviation is reported with its upper and lower 95% confidence limits (S_u and S_l) determined from tables presented by *Cox* [1969*b*]. It should be noted that due to the episodic nature of volcanic activity and the lack of stratigraphic control, the N sites may not represent independent events. Since this is probably the case, the confidence limits are actually larger than reported.

Secular Variation Models

At present, five models have been introduced to represent secular variation and explain its obvious dependence on latitude. *Irving and Ward* [1964] proposed model A with secular variation due to a nondipole component of random direction but constant intensity, and no dipole wobble. Model B [*Creer et al.*, 1959] is a one-parameter model with a constant secular variation from the wobble of the main dipole and no nondipole effects. Model C [*Cox*, 1962; *Creer*, 1962] combines the nondipole and dipole wobble components, so that total angular dispersion (S_t^2) is due to the sum of the dispersion due to the dipole wobble (S_d^2) and the dispersion of the nondipole components (S_n^2). This latter term is evaluated by using averaged data from the 1965 international geomagnetic reference field. Model D [*Cox*, 1970] is similar to model C except that S_n is calculated mathematically rather than from present-day nondipole magnetic data. A fifth model, E, which defines a band of secular variation rather than a single curve, has been proposed by *Baag and Helsley* [1974]. The lower margin of the band is the same as model C, while the upper limit is determined from the square of the sum of deviation due to dipole wobble and the deviation due to the nondipole field.

Models B, C, and E are shown in Figure 6. Numerical values for the curves may be found in the works by *Cox* [1970] and *Baag and Helsley* [1974]. The circles with inscribed crosses are

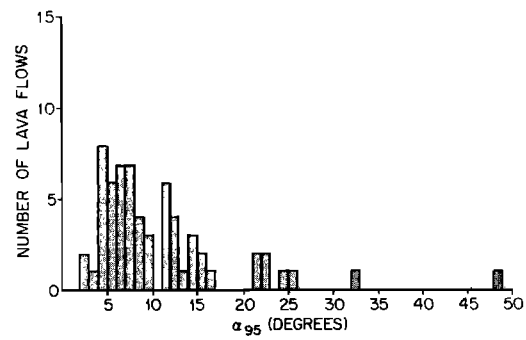


Fig. 5. Histogram of α_{95} values for 63 Terevaka flows.

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TABLE 2. Secular Variation Statistics

	S_T	K_T	K_W	K_A	K_F	S_F	S_u	S_l
Set A								
with respect to VGP	13.07	38.40	32.08	2100	56.27	10.80	13.06	9.19
with respect to geographic pole	13.09	38.31			56.07	10.82	13.09	9.23
Set B								
with respect to VGP	15.79	26.33	19.57	2100	40.44	12.74	14.75	11.22
with respect to geographic pole	15.80	26.29			40.34	12.75	14.78	11.21

S_T represents total dispersion of VGP's; K_T, total precision parameter; K_W, within-site precision parameter; K_A, precision parameter due to ancient local anomalies [*Doell*, 1970]; K_F, precision parameter for VGP dispersion due to ancient secular variation; S_F, corresponding dispersion; and S_u and S_l, upper and lower confidence limits at 95% level [*Cox*, 1969*b*]. For discussion and formulas for parameter determinations, see *Ellwood et al.* [1973].

secular variation data points determined by *Doell and Cox* [1972] for Brunhes age igneous rocks. The open circles are Brunhes age data points presented by other authors, listed in the figure heading. Easter Island is shown to lie slightly below the curve of model C and the lower limit of model E, although it is not as low as the Hawaiian data. It appears that the central Pacific secular variation low discussed by *Doell and Cox* [1971, 1972] may extend into the southeast Pacific.

COMMENTS

This study presents an additional data set suitable for examining the latitude dependence of secular variation, the extent of the Pacific secular variation low, and mean geomagnetic pole for the Brunhes epoch. The data do not provide significant constraints by themselves for choosing between model C or D and model E for observed secular variation. A specific intent of model E is to explain the observation that dispersion of the magnetic field is greater at high latitudes than the theoretical dispersion based on constant average nondipole intensity at all latitudes [*Baag and Helsley*, 1974]. The data sets cited and/or recalculated by *Baag and Helsley* [1974] primarily have angular dispersion values greater than those predicted by model C. The low value for secular variation on Easter Island is compatible with both models.

An original aim of this study was to investigate the southeastern Pacific for an extension of the central Pacific secular variation low. Historical data [*Doell and Cox*, 1972, Figures 1–4] suggest that Easter Island is located near the edge of the anomalous region that has persisted in the central Pacific for at least the last 0.7 m.y. *Doell and Cox* [1972] postulate inhomogeneities in the lower mantle or undulations of the core-mantle boundary as possible causes of the subdued nondipole field. As can be seen from Figure 6, Easter Island lies above the dipole wobble component of secular variation but below the lower limit for all proposed models. Thus it appears that the Pacific secular variation low extends into the southeast but is not as pronounced as it is around Hawaii. The Galapagos Islands in the eastern Pacific appear to be unaffected by the low, as the angular dispersion value is similar to that predicted by the models.

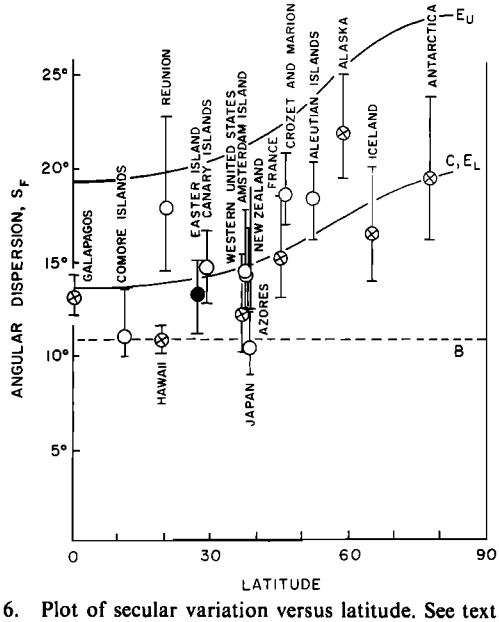


Fig. 6. Plot of secular variation versus latitude. See text for discussion of curves. The circles with inscribed crosses are from *Doell and Cox* [1972]. Other data are from Comore, Amsterdam [*Watkins and Nougier*, 1973]; Reunion [*Watkins*, 1973]; Canary, Azores [*Ellwood et al.*, 1973]; Japan [*Ozima and Aoki*, 1972]; Crozet and Marion [*Amerigian et al.*, 1974]; and Aleutians [*Bingham and Stone*, 1972]. All studies are on Brunhes age rocks except for the Aleutian Islands, which include upper Tertiary rocks.

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